

WILLIAM G. LEWIS

AI Infrastructure & LLMops Engineer | Classified AI Platforms | AWS GovCloud | RAG

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Active Top Secret (TS) Clearance | SCI Eligible | IT-1 | Security+ (DoD 8140/8570 IAT Level II) | AWS GovCloud IL2-IL6

PROFESSIONAL SUMMARY

AI Infrastructure and LLMops Engineer with 7+ years of cloud and platform engineering experience and hands-on ownership of private large language model inference, retrieval-augmented generation, GPU-backed compute, and security telemetry in air-gapped AWS GovCloud environments. Pioneered a program's first operational AI security capability and designed an executive proof of concept for classified AI-enabled investigation. Combines model-serving infrastructure with Terraform, GitLab CI/CD, OpenSearch, Docker, observability, security controls, and operational documentation.

CORE TECHNICAL COMPETENCIES

AI Infrastructure: Local LLM inference, Ollama, GPU EC2, NVIDIA T4, Open WebUI, FastAPI, AWS Bedrock

RAG & Search: Retrieval-Augmented Generation, OpenSearch retrieval, query-generation pipelines, OpenSearch Dashboards

Cloud & IaC: AWS GovCloud, EC2, VPC, IAM, Lambda, API Gateway, DynamoDB, S3, ECS, RDS, Terraform, CloudFormation, Ansible

Platform Delivery: GitLab CI/CD, Jenkins, Docker, Amazon ECR, Kubernetes/EKS, GitOps, cross-domain code promotion

Observability: OpenSearch, Fluent Bit, CloudWatch, Lambda SIEM pipelines, Dynatrace

Languages & APIs: Python, Bash, PowerShell, Boto3, RESTful APIs, YAML, JSON

SELECTED AI INFRASTRUCTURE PROJECTS

- **CNAP Log Analyst Agent:** Pioneered and deployed a production GPU-backed local LLM inference platform in an air-gapped DoD environment with zero commercial cloud dependency; integrated OpenSearch RAG for natural-language investigation of Palo Alto and AppGate security data, resolved a production credential-rotation defect, and delivered operational documentation for classified handoff.
- **CNAP AI SIEM Chatbot:** Designed the architecture and proof of concept for a classified AI security-intelligence chatbot; defined the LLM integration layer, OpenSearch query-generation pipeline, and GovCloud deployment model, then presented the design to C1 Cohort leadership and secured go/no-go approval for AI VPC expansion.
- **AI Cloud Sentiment Analysis & Summarizer:** Built a serverless transcript-analysis platform with AWS Bedrock, Lambda, DynamoDB, and API Gateway, generating structured AI summaries through a REST API and reducing latency by 50% through architecture optimization.

PROFESSIONAL EXPERIENCE

Lead Cloud & Platform Engineer, IL6 Technical Lead (Platform & Security Architecture) | SAIC / Oteemo, CNAP Program, Gunter AFB | Nov 2023 - Present

AWS GovCloud (IL4/IL5/IL6), AppGate SDP, Palo Alto NGFW (PAN-OS), Terraform, GitLab CI/CD, OpenSearch, Fluent Bit, Ollama, NIST RMF

- Pioneered the program's first AI-enabled security operations capability: deployed a local LLM inference platform in a classified, air-gapped environment, enabling natural-language investigation against live security log data, a capability with no prior program equivalent.
- Designed and owned the program's classified SIEM pipeline from zero (Fluent Bit to OpenSearch) with IAM-based authentication, index architecture, and operational dashboards, delivering the team's first end-to-end log visibility.
- Containerized and delivered hardened application stacks with Docker and Amazon ECR into the air-gapped IL6 environment in compliance with STIG and CIS Benchmark controls.
- Architected a high-side CI/CD pipeline for diode-constrained Infrastructure as Code delivery, engineering a versioned S3 cross-domain staging workflow that solved a classified automation problem with no off-the-shelf solution.
- Standardized Terraform Infrastructure as Code across the classified stack through per-component deployment scripts with CIDR enforcement and environment-variable injection, reducing estimated deployment errors by 60-70% and eliminating manual provisioning overhead.
- Served as primary technical owner of the CNAP IL6 platform, directing engineering across Zero Trust networking, NGFW operations, CI/CD automation, SIEM observability, and ATO compliance from initial IATT through production hardening and customer onboarding.
- Led AppGate Software-Defined Perimeter deployment and Zero Trust hardening across TEST and PROD, configuring entitlement governance, CIS Benchmark controls, and SCAP-validated posture; established the operational Zero Trust baseline for SAOC customer onboarding.
- Directed IATT Test Plan execution across five sprint cycles, coordinating with ISSM and SCA-R, authoring the full ATO documentation suite (IRP, ConMon, VMP, CMP, SAR), and closing POA&Ms across ACAS, STIG, and AWS Foundations to deliver a compliant RMF package.

- Governed the Palo Alto NGFW lifecycle, including licensing, PAN-OS version upgrades (10.2.8 to 11.0.3), patching, and policy hardening across TEST and PROD simultaneously, maintaining security compliance with zero unplanned downtime.
- Established the IL4/IL5 customer onboarding framework: assessed process gaps, created the onboarding epic, and delivered a repeatable architecture enabling mission-customer expansion on the CNAP platform.

Senior AWS Cloud Consultant | Baker Tilly | Dec 2021 - Jun 2023

AWS Commercial, Terraform, CloudWatch, Docker, RDS, Multi-Region Migration

- Led a multi-region cloud migration from Rackspace to AWS using modular Terraform stacks, standardizing provisioning patterns and reducing manual configuration overhead across customer environments.
- Built cross-account RDS backup automation and Docker image pipelines for ARM and x86, improving data resilience and standardizing secure CI/CD delivery.
- Designed proactive monitoring through CloudWatch Canaries and Amazon SNS-to-Microsoft Teams integration, reducing mean time to detection for infrastructure incidents.

Senior Cloud Engineer | SAIC | Aug 2020 - Dec 2023

AWS, CloudFormation, Ansible, Jenkins, Git, PowerShell, CloudWatch, Elastic Beanstalk, EC2, ACAS, F5, Solaris 10

- Drove Tier 2 and Tier 3 support triage by monitoring and troubleshooting problems, implementing scheduled and emergency patches, running ACAS scans, and executing engineering plans for application- and platform-level updates.
- Deployed Infrastructure as Code changes across 24 AWS accounts and DEV/TEST/PROD environments using CloudFormation and Ansible, with Git for version control and Jenkins for continuous integration and delivery.
- Developed an automated Active Directory user report for key government stakeholders using Jenkins, PowerShell, and Ansible.
- Automated application availability validation using PowerShell, Ansible, and Jenkins, supporting a 99.99999% system uptime service-level agreement.
- Created a CloudWatch uptime-monitoring dashboard for Elastic Beanstalk and EC2 performance, improving resource visibility and response times.
- Led the TLS 1.0-to-1.2 transition across DEV/TEST/PROD environments, remediating F5 firewall findings and maintaining DoD cybersecurity compliance.
- Administered patching and quality analysis for Solaris 10 systems through a complex multi-step procedure, maintaining compliance and availability for mission-critical ESB legacy systems.

RHEL Cloud System Administrator | Direct Viz Solutions | Oct 2019 - Feb 2020

Red Hat Enterprise Linux, Cloud Infrastructure, Middleware, OS Patching, Performance Monitoring

- Managed cloud infrastructure services, applying middleware and operating-system updates while identifying and resolving performance inefficiencies.
- Developed, configured, and supported cloud platforms and infrastructure for optimal performance and reliability.

Cloud System Integrator | Leidos | Jun 2018 - Oct 2019

Cloud Migration, Configuration Management Database, SSL/TLS Monitoring, Application Support, 24/7 Operations

- Collaborated with technical leads to support application migrations and provided 24/7 operational support to maintain customer service and system availability.
- Built a Configuration Management Database to track resources, reduce downtime, and improve patching efficiency.
- Developed a system to monitor expiring SSL certificates, maintaining continuous compliance and supporting a 99.99999% availability service-level agreement.

CERTIFICATIONS & CLEARANCE

- Active Top Secret (TS) Security Clearance, SCI Eligible, IT-1
- AWS Certified Solutions Architect - Associate (2022)
- AWS Certified SysOps Administrator - Associate (2023)
- CompTIA Security+ CE (active), DoD 8140/8570 IAT Level II compliant
- AWS Certified AI Practitioner (in progress)

EDUCATION

Bachelor of Science in Computer Science: Alabama State University, Cum Laude | Graduated Dec 2018

TECHNICAL DOCUMENTATION & PUBLICATIONS

- Authored documentation defining Infrastructure as Code deployment best practices for GitHub, environment-redeployment procedures, and branching strategies for Infrastructure as Code environments.
- Published articles on cloud infrastructure management and optimization techniques.

AFFILIATIONS

- Alpha Phi Alpha Fraternity, Inc. (2016-Present)
- Scouting America (2023-Present)
- National Society of Black Engineers (2018-Present)